CLOSED IMMERSIONS OF BASES INDUCE CLOSED IMMERSIONS OF COH STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 511-517 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 519-552, the complete author proof (32 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 12 of queue_moduli.json (pb=1414, stmt 511-517, proof 519-552).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DN9: sources/stacks-moduli/pinned-original-source.tex, source lines 425–430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 083M: sources/stacks-moduli/pinned-original-source.tex, source lines 552–552. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04CJ: sources/stacks-moduli/pinned-original-source.tex, source lines 552–552. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let B be an algebraic space. Let $\pi : X \to Y$ be a closed immersion of algebraic spaces which are separated and of finite presentation over B. Then the morphism $\mathcal{C}oh_{X/B} \to \mathcal{C}oh_{Y/B}$ of Lemma 0DN9 is a closed immersion.*

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $\mathcal{I} \subset \mathcal{O}_Y$ be the sheaf of ideals cutting out X as a closed subspace of Y. Recall that π_* induces an equivalence between the category of quasi-coherent $\mathcal{O}_X$-modules and the category of quasi-coherent $\mathcal{O}_Y$-modules annihilated by $\mathcal{I}$, see Morphisms of Spaces, Lemma 04CJ. The same, mutatis mutandis, is true after base by $T \to B$ with $\mathcal{I}$ replaced by the ideal sheaf $\mathcal{I}_T = \mathrm{Im}((Y_T \to Y)^*\mathcal{I} \to \mathcal{O}_{Y_T})$. Analyzing the proof of Lemma 0DN9 we find that the essential image of $\mathcal{C}oh_{X/B} \to \mathcal{C}oh_{Y/B}$ is exactly the objects $\xi = (T \to B, \mathcal{F})$ where $\mathcal{F}$ is annihilated by $\mathcal{I}_T$. In other words, ξ is in the essential image if and only if the multiplication map

$$\mathcal{F} \otimes_{\mathcal{O}_{Y_T}} (Y_T \to Y)^*\mathcal{I} \longrightarrow \mathcal{F}$$

is zero and similarly after any further base change $T' \to T$. Note that

$$(Y_{T'} \to Y_T)^*(\mathcal{F} \otimes_{\mathcal{O}_{Y_T}} (Y_T \to Y)^*\mathcal{I}) = (Y_{T'} \to Y_T)^*\mathcal{F} \otimes_{\mathcal{O}_{Y_{T'}}} (Y_{T'} \to Y)^*\mathcal{I})$$

Hence the vanishing of the multiplication map on T' is representable by a closed subspace of T by Flatness on Spaces, Lemma 083M. $\square$